



Effects of sensors

Study on the different sensors and computation systems in AVs, and how they effect vehicles over their lifetime. <https://digit.in/oct19-63>



Cooperative Driving

An exploration of security threats of state of the art autonomous driving systems to design a reference architecture. <https://digit.in/oct19-64>

cles are automated but not electrified, emissions will go up by 50 percent, by 2050. However, if the vehicles are electrified and automated, greenhouse gas emissions are projected to reduce by 80 percent. The production of the electricity itself could have a negative impact on the environment.

Broadly speaking, there are a few factors that are expected to contribute to fewer emissions. Instead of mostly one person per vehicle, ridesharing is expected to become increasingly common. The easy availability of shared mobility on demand is expected to be a major factor contributing to reduced emissions. On demand mobility is also expected to reduce the number of vehicles owned by individuals. This would be helped by emerging ownership models such as car sharing and subscriptions. Platooning of vehicles and co-ordination between them is expected to ease congestion, get rid of stop-and-go traffic, and improve

ably seen an increase in use. More people would be able to use vehicles, those that are unable to use vehicles on their own. These include the elderly, the infirm, those with disabilities and children. There are indications that travel demand would increase with automation. At the same time, because of the reduced pressure on driving, commuting distance can increase. People may find it more comfortable to spend more time in vehicles, as they no longer have to keep their eyes on the road. The physical sensors and systems needed for autonomous cars increase the lifetime energy consumption of vehicles. They also contribute to the weight of the vehicles and increase the drag. Also, there may be times when autonomous vehicles are travelling to places without anyone in them. These can be autonomous taxis navigating to pick up locations, or long distance autonomous buses with a low number of filled seats. For all

of autonomous vehicles stand to change urban landscapes. The factors affecting cities are the same as the environment. If more people choose ridesharing options, then there will be a reduced demand for parking spaces. In an ideal scenario, most vehicles would be on the road and occupied, instead of waiting for their owners in parking spaces. Cities may generate far fewer revenues from parking spaces. Also, the real estate prices may get readjusted because more space would be made available for other kinds of uses. A potential new source of revenue could be pick up and drop off points, which would require appropriate designs as well, especially in crowded areas. Access to these points may be charged by cities, for example.

The designs of the roads are also likely to change once all the vehicles are fully autonomous. Instead of dedicated lanes for buses and trucks for example, all the vehicles could simply be programmed to leave a lane open only when a bus is passing by. It is in fact possible for policy makers to use these technologies to make sure that autonomous vehicles are ultimately beneficial. For example, the movement of vehicles with single occupancy can be restricted, or higher tolls extracted.

Historically, one of the biggest factors affecting the size of cities, has been the amount of time it takes to go from the periphery, to the urban centers. With the introduction of vehicles, expressways and high speed transit systems, the sizes of cities have also grown exponentially. With the rollout of autonomy, the amount of time people are willing to spend in commuting might increase. So far, irrespective of the mode of travel, the average duration around the world has been around 30 minutes. That means, that it takes about 30 minutes to reach the urban centres from the outskirts of the villages. If the passengers of the future autonomous vehicles are more tolerant of long commute times, then it is very likely that metropolitan foot-



A gridlock with regular traffic in Manhattan

travel times, all of which together contribute towards reduced emissions. Because of the increased safety offered by autonomous vehicles, the weight of the vehicles can be reduced by removing the safety systems.

However, all these positives mean that people have a strong incentive to increase vehicle use. Researchers refer to these category of potential pitfalls as the "rebound effects". Any technology that saves time and money has invari-

the factors that could lead to reduced emissions, there are an equal number of factors that could lead to an increase. At this point in time, researchers cannot guess even if the introduction of autonomous vehicles will have a net positive or net negative impact, when it comes to the environment.

CITIES OF THE FUTURE

Closely related to the environmental impact, is how the introduction